The Thomson problem for N = 10

P. Graham
https://orcid.org/0009-0007-8493-1505

October 24, 2026

Version 5

Abstract

For ten distinct points on the unit sphere, we prove that the bicapped square antiprism uniquely minimizes Coulomb energy, up to orthogonal maps and relabeling. This settles the ten-point case of Thomson's problem, associated with his 1904 atomic model. The height is the unique critical point in an exact rational interval. The proof first confines all competitors of no greater energy to one inner-product pattern, aligns them with the candidate, and applies a quantitative local energy bound.

The proofs, the Lean formalization, the certificate checkers and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

1 Statement and candidate

This section specifies the configuration and the exact optimization problem. Every competitor consists of ten distinct unit vectors. For an injective map $x : \{0, \dots, 9\} \to S^2 \subset \mathbb{R}^3$, put

$$\mathcal{E}(x) = \sum_{0 \le i < j \le 9} \frac{1}{\|x_i - x_j\|}.$$

We call two configurations congruent when one is obtained from the other by an element of $O(3)$ and a permutation of the ten indices; reflections are allowed.

For $0 < h < 1$, write $r = \sqrt{1 - h^2}$ and $q = r/\sqrt{2}$. Define $p(h)$ by

$$\begin{aligned} &p_0 = (0, 0, 1), && p_1 = (0, 0, -1), \\ &p_2 = (r, 0, h), && p_3 = (0, r, h), && p_4 = (-r, 0, h), && p_5 = (0, -r, h), \\ &p_6 = (q, q, -h), && p_7 = (-q, q, -h), && p_8 = (-q, -q, -h), && p_9 = (q, -q, -h). \end{aligned}$$

The family energy is $F(h) = \mathcal{E}(p(h))$, with the explicit expression in (1). Write $S(h) = F'(h)$. Let $[a, b]$ be the exact rational interval in Appendix D.1, and define h_* by $S(h_*) = 0$ and $h_* \in [a, b]$. Its decimal begins $0.422687424043986\ldots$.

Theorem 1. *The parameter h_* exists and is unique in $[a, b]$. The configuration $p = p(h_*)$ consists of ten distinct points on S^2. For every injective $x : \{0, \dots, 9\} \to S^2$,*

$$\mathcal{E}(p) \le \mathcal{E}(x), \qquad \mathcal{E}(x) = \mathcal{E}(p) \iff \exists Q \in O(3),\ \exists \sigma \in S_{10},\ x_i = Q p_{\sigma(i)} \text{ for all } i.$$

No symmetry is assumed of competitors. The candidate has two poles and two squares at opposite heights, rotated by $\pi/4$ (Figure 1). Optimizing its height alone would not exclude asymmetric competitors.

Context and contribution. The historical reference is Thomson's 1904 paper [1, p. 237], whose opening specifies the "motion of a ring of n negatively electrified particles". For the general question, Cohn and Woo [4, Section 1.1, arXiv v2] use the phrase "potential energy minimization". Here the metric space is the unit sphere and the potential is inverse distance. Our immediate source is Aravind Karthigeyan's MIT-licensed nine-point project [2, README, revision `168ff0c`], which states "global optimality and uniqueness for nine distinct points". We adapt its formal code and proof design. The new inputs are the ten-point energy, interval-classification and Hessian certificates. The hardest step is the three-point energy certificate: its error must be small enough to force geometric localization. The interval centered at -1 contains the antipodal pole pair; a five-point counting argument supplies the linearly independent triple needed for classification.

Proof overview. For a competitor of energy at most $\mathcal{E}(p)$, a two-point bound first gives $\langle x_i, x_j \rangle < 13/20$ for distinct indices. A degree-16 polynomial minorant and a rational three-point certificate bound the total pairwise slack (pair energy minus its polynomial lower bound) by $L < 10^{-30}$. Outside seven intervals of radius $\delta = 10^{-10}$ around the candidate's inner products, even one pair's slack would be at least 10^{-30}. A finite Gram-determinant classification forces a common interval pattern after relabeling. Orthogonal least-squares alignment then gives total displacement $\rho = (\sum_i \|y_i - p_i\|^2)^{1/2} \le 10000\delta = 10^{-6}$ for the aligned configuration y. Full spherical stationarity, a constrained-Hessian bound and a cubic remainder estimate give $\mathcal{E}(y) - \mathcal{E}(p) \ge \rho^2/200$ for an aligned configuration with $\rho \le 10^{-3}$. Hence $\rho = 0$. Appendices B and C give every reduction.

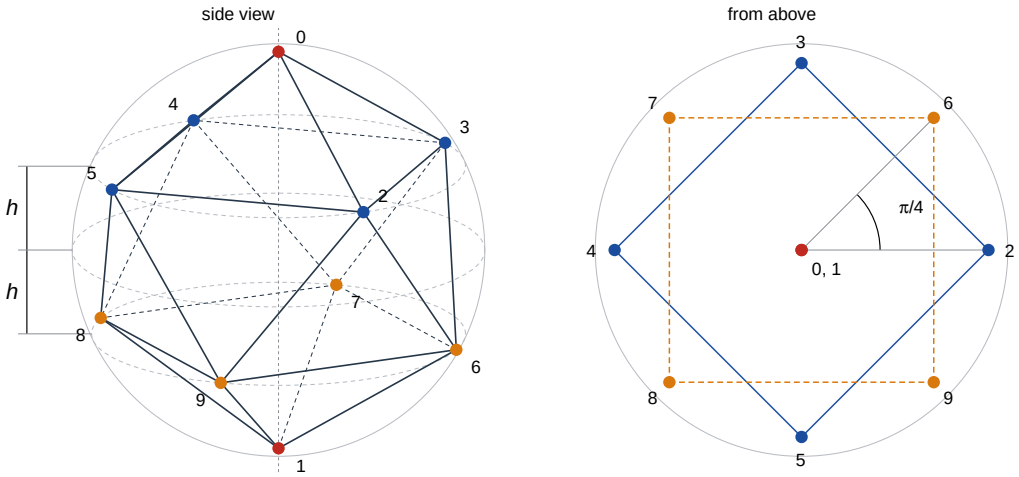


Figure 1: The candidate $p(h_*)$. Left: a side view of the ten points and the edges of their convex hull, hidden edges dashed. The poles p_0, p_1 are red, the upper square $p_2, \dots, p_5$ at height $h = h_*$ is blue, and the lower square $p_6, \dots, p_9$ at height $-h$ is orange; the dashed ellipses are the equator and the two circles of latitude. Right: the view from above, in which the lower square (dashed) is the upper square rotated by $\pi/4$. A label i marks p_i.

Checked scope and reproduction. Lean proves the root assertion, candidate admissibility, global optimality and the entire equality equivalence from the definitions; all certificate obligations are discharged. The trust base is the Lean kernel, the standard axioms `propext`, `Classical.choice`, `Quot.sound`, and the match between the formal definitions and the statement above. There is no native-computation axiom. The independent Python implementation checks the same rational three-point certificate and rejects its corruptions; it does not independently prove the geometric or local arguments. Its decimal energy enclosure and figure check are supplementary, outside the Lean theorem. The figure is drawn from the Lean definition of the candidate by `checks/draw_antiprism.py`, which checks that the drawn points are the candidate's; it is not part of the proof. Appendix D specifies these boundaries.

With elan, Git, gzip and Python 3.10 or later installed, run from the result folder:

```
cd lean
lake exe cache get
lake build
cd ..
python3 -I checks/check.py
```

The toolchain and dependencies are pinned. The guides in `lean/README.md` and `checks/README.md` give memory requirements and the certificate inputs.

A Candidate details

This appendix records the family energy and the quantitative parameter facts used in Theorem 1. These coordinates are preserved by rotation through $\pi/4$ about the polar axis followed by reflection in the equatorial plane, and by reflection in the vertical plane through p_2 and the poles. The 16-element group generated by these maps is conventionally denoted D_{4d}.

The 45 unordered pairs consist of one pole pair, eight neighboring and four opposite pairs within the squares, two classes of eight pairs between the squares, and sixteen pole–square pairs. Grouping them gives

$$F(h) = \frac{1}{2} + \frac{4\sqrt{2}+2}{\sqrt{1-h^2}} + \frac{8}{\sqrt{2-\sqrt{2}+(2+\sqrt{2})h^2}} + \frac{8}{\sqrt{2+\sqrt{2}+(2-\sqrt{2})h^2}} + \frac{8}{\sqrt{2-2h}} + \frac{8}{\sqrt{2+2h}}. \tag{1}$$

The terms appear in that order: pole pair, pairs within the squares, the two between-square classes, and the two pole–square classes. Write $S(h) = F'(h)$ for the derivative of this energy with respect to height. Its explicit formula is given in Appendix D.1. Let $[a, b]$ be the exact rational interval specified in Appendix D.1. It satisfies $0 < a < b < 1$ and $b - a = 1/(2D) < 2^{-300}$, where the integer denominator D is given there. For orientation, this narrow interval lies inside

$$[0.42268742404398, 0.42268742404400].$$

The derivative S is continuous and strictly increasing on $[a, b]$, and its endpoint signs satisfy $S(a) < 0 < S(b)$. The intermediate value theorem and strict monotonicity therefore give a unique zero in this interval. Define h_* by $S(h_*) = 0$ and $h_* \in [a, b]$. Its decimal begins $0.422687424043986\ldots$. The root existence and uniqueness in this specified interval are proved in Lean. This interval selects the candidate; the global theorem still compares it with every admissible configuration, including all other heights.

Here and below, admissibility means precisely unit norm and injectivity. The candidate is admissible: in the coordinates above, unit norm follows from $r^2 = 1 - h_*^2$ and $2q^2 = r^2$. The two poles and the two squares have distinct heights; within each square the four planar coordinates are distinct because $r, q > 0$. Thus the ten points are distinct. Appendix D.2 identifies the formal statements of both the optimization theorem and this candidate-validity assertion.

The seven distinct inner products of the candidate are

$$-1, \quad h_*, \quad -h_*, \quad h_*^2, \quad 2h_*^2 - 1, \quad \frac{1-h_*^2}{\sqrt{2}} - h_*^2, \quad -\frac{1-h_*^2}{\sqrt{2}} - h_*^2.$$

Remark (energy enclosure). A separate rigorous evaluation of the candidate energy gives

$$32.716949460147 < \mathcal{E}(p) < 32.716949460149,$$

so the energy starts $32.71694946\ldots$. This rigorous rational-interval computation is separate from the formal theorem. Appendix D.8 gives the independent Python calculation and its exact inputs.

B From energy bounds to a common interval pattern

We now show that every admissible competitor of energy at most the candidate's has nearly the same matrix of pairwise inner products, after relabeling. Let $f(t) = (2-2t)^{-1/2}$. For unit vectors,

the pair energy is $f(\langle x_i, x_j\rangle)$. Throughout this section, $p = p(h_*)$ and x is an arbitrary admissible configuration satisfying $\mathcal{E}(x) \le \mathcal{E}(p)$.

B.1 Energy bounds force seven windows

We bound the energy lost by replacing the pair potential with a polynomial; this loss will force every pair into one of seven short intervals.

Pair separation. Rational interval evaluation gives $\mathcal{E}(p) \le 32717/1000$ (Appendix D.5), so this is also an energy cap for the competitor. Let g be the quartic polynomial specified in Appendix D.3. It lies below f on the whole interval $[-1, 1)$ and has nonnegative coefficients in the Legendre basis. The spherical-harmonic addition identity therefore expresses the sum of $g(\langle x_i, x_j\rangle)$ over all ordered pairs, including the diagonal, as a constant plus nonnegative weighted squares. Each square is the square of the sum, over the ten points, of a fixed spherical harmonic of degree one through four. Subtracting the diagonal contribution and dividing by two gives a lower bound of $13111005315721/401353681050 > 32.666$ for the sum over unordered distinct pairs. If any pair had inner product at least $13/20$, its energy minus g would exceed $51/1000$. All other pair differences are nonnegative. Adding $51/1000$ to the exact polynomial lower bound exceeds $32717/1000$ by more than 0.00096, a contradiction. Thus $\langle x_i, x_j\rangle < 13/20$ for every $i \ne j$.

Three-point bound. The next bound uses a fixed rational polynomial and a finite set of rational matrices; together these data form the *three-point certificate*. Define $m(t) = \sum_{j=0}^{16} c_j t^j - 2^{-130}$, where the coefficients c_j are `hCoeff` in `lean/ThomsonN10/Data.lean`. The downward shift ensures that m lies below f throughout $[-1, 13/20]$. Let U be the fixed rational `certU` in the same file. Interval evaluation of the candidate gives $\mathcal{E}(p) \le U$, where $U \approx 32.716949460147588154555705751969994$. Appendix D.5 gives its exact value.

For three unit vectors x_i, x_j, x_k, use the argument convention $u = \langle x_i, x_j\rangle$, $v = \langle x_i, x_k\rangle$ and $t = \langle x_j, x_k\rangle$. The matrices specify two polynomials K and W. Define the residual polynomial $\mathcal{R}$ by

$$\frac{m(u) + m(v) + m(t)}{3} - \frac{U - 45 \cdot 2^{-130}}{45} = K(u, v, t) + W(u, v, t) + \mathcal{R}(u, v, t).$$

The content of the certificate is threefold. First, the sum of K over all ordered triples of distinct indices is nonnegative for every configuration of ten unit vectors. For exact prior statements see Bachoc–Vallentin [3, Corollary 3.5, arXiv v4] and Cohn–Woo [4, Theorem 5, arXiv v2]. The repeated-index corrections are specified in Appendix D.4. Second, W is a sum of weighted polynomial squares, and is nonnegative on the *capped Gram domain* $u, v, t \in [-1, 13/20]$, $1 + 2uvt - u^2 - v^2 - t^2 \ge 0$. The six weights are listed in Appendix D.5. Third, the sum $\|\mathcal{R}\|_1$ of the absolute values of the residual's rational coefficients is tiny: $\|\mathcal{R}\|_1 \approx 1.1673222012517146 \cdot 10^{-56}$. Every residual monomial has absolute value at most one on this domain, so this coefficient sum bounds the residual pointwise.

Put $L = 45(2^{-130} + \|\mathcal{R}\|_1)$; numerically, $L \approx 3.306077861687684 \cdot 10^{-38} < 10^{-30}$. These decimals describe the exact rational data, and are not used as proof inputs. Averaging the defining identity over distinct triples and using the two positivity properties gives the following bound.

Lemma 1 (Certified energy bound)**.** *For the fixed polynomial m and constants U, L just specified, every configuration of ten unit vectors with $\langle x_i, x_j\rangle \le 13/20$ for all $i \ne j$ satisfies*

$$\mathcal{E}(p) \le U \le 32717/1000, \qquad U - L \le \sum_{i<j} m(\langle x_i, x_j\rangle), \qquad L < 10^{-30}.$$

The candidate inequality in the display is the separately established interval bound above. For the pair-sum inequality, the unit-vector hypotheses put every triple in the Gram domain, while the pair cap gives $u, v, t \le 13/20$, where all weights of W are nonnegative. Appendix D.5 gives the triple-to-pair averaging and explains the factor 45 in L. The much weaker separation bound used Delsarte–Yudin linear programming; here the error is small enough to force every pair into a candidate window.

For each pair, call the nonnegative difference between its true energy and its polynomial lower bound its *slack*. Their total is at most L, because the energy is at most that of the candidate:

$$\sum_{i<j}(f-m)(\langle x_i, x_j\rangle) = \mathcal{E}(x) - \sum_{i<j} m(\langle x_i, x_j\rangle) \le \mathcal{E}(p) - (U-L) \le L.$$

Slack outside the windows. A *window* is an open interval of radius $\delta = 10^{-10}$ centered at one of the seven candidate inner products in Appendix A. The slack satisfies $f(t) - m(t) \ge 10^{-30}$ for every $t \in [-1, 13/20]$ outside all seven windows. Since $L < 10^{-30}$, even one pair outside the windows would exceed the total available slack. Every pair therefore belongs to a window.

B.2 The window pattern is the candidate's pattern

We now classify which assignments of these seven windows to the pairs can arise from ten unit vectors in three dimensions. Number the seven candidate inner products by the colors zero through six, as in the table below. Assign a color to each pair according to the window containing its inner product. For example, the candidate's pole pair receives the color of the window centered at -1. The windows are disjoint, so each pair has just one color.

Color	Candidate inner product	Number of pairs
0	-1	1
1	h_*	8
2	$-h_*$	8
3	h_*^2	8
4	$2h_*^2 - 1$	4
5	$(1-h_*^2)/\sqrt{2} - h_*^2$	8
6	$-(1-h_*^2)/\sqrt{2} - h_*^2$	8

An *anchor triple* consists of three points whose three-by-three Gram matrix has determinant greater than $1/100000$. Such a triple is linearly independent. For each sorted triple of pair colors, evaluate the three-by-three Gram determinant over rational outer enclosures of the three windows. The resulting *determinant interval* contains every determinant that those colors can realize. An anchor *type* is a sorted triple of pair colors. A type is *eligible* when this interval's upper endpoint is greater than the threshold; every actual anchor triple has an eligible type.

An anchor triple must exist. The determinant intervals show that a triangle with determinant at most the threshold can have only a color permutation of $(0, 1, 2)$, $(1, 1, 4)$ or $(2, 2, 4)$. Each such triangle has exactly one edge of color zero or four. If every triangle among the five configuration points with indices zero through four had one of these types, counting these marked edges over the ten triangles would give ten. But each edge belongs to three triangles, so the same count must be divisible by three. This contradiction supplies an anchor triple. The subsequent enumeration treats all 43 eligible anchor types. Reorder the three anchor points so that the colors of their pairs (first, second), (first, third), (second, third) are nondecreasing. Permuting three vertices permits

every permutation of their three edges, so such an order always exists. The node coordinates below use this fixed order. For example, the candidate triple p_2, p_3, p_6 has ordered pair colors $(3, 5, 5)$.

Each of the remaining seven points determines a *node*: the three colors of its pairs with the three anchor points. The four-by-four Gram determinant of this point and the anchor points vanishes because four vectors lie in three dimensions; interval evaluation removes nodes for which this is impossible.

For two remaining points, form a four-by-four matrix with rows indexed by the three anchor points followed by the first point, and columns indexed by the same three anchor points followed by the second point. Each entry is the inner product of its row-index and column-index vectors. This is their *cross-Gram matrix*. It has rank at most three, so its determinant vanishes. Two nodes are *compatible* with a proposed color between them when interval evaluation of this determinant contains zero. The nodes specify the windows for inner products with the anchor points, and the proposed color specifies the window for the one remaining entry.

Join two nodes when at least one color survives the pair test, retaining the list of all surviving colors on that edge. A *clique* is a set of seven nodes with an edge between every two. The seven remaining points of an actual configuration give such a clique. They give distinct nodes because the pair test excludes all seven colors when the two node indices are equal; that test still allows the two points' inner products with the anchors to vary independently within the same windows.

The enumeration checks every clique in every eligible anchor type. For each clique, its witness is a bijection from the three anchor points and seven nodes to the candidate's ten indices. The witness preserves the anchor and node colors and checks *every surviving color* on each clique edge against the candidate color. Thus no separate choice of edge colors is needed: all colors that an actual configuration could use are forced to agree with the same candidate labeling. The nine witnesses correspond to nine enumerated node subsets with their anchor types; all give the same candidate pattern up to relabeling.

Classification. The finite classification proves that there exists a permutation τ for which

$$|\langle x_{\tau(i)}, x_{\tau(j)}\rangle - \langle p_i, p_j\rangle| \le \delta \quad \text{for all } i, j. \tag{2}$$

Thus each pair has been assigned the same window as the corresponding candidate pair. The permutation τ labels the competitor by the candidate indices. The conclusion fixes an interval pattern and bounds every Gram-matrix entry; it does not yet say that the inner products are equal. The certificate counts and its formal soundness statement are given in Appendix D.5.

C Alignment and exact local rigidity

This section proves that a competitor of no greater energy satisfying (2) is congruent to the candidate. We first align it with the candidate and then show that any remaining displacement raises energy.

C.1 From inner products to point displacements

We convert the bounds on inner products into a small displacement from the candidate. Among orthogonal maps, choose Q minimizing the sum of squared displacements from p, and put

$$y_i = Q x_{\tau(i)}, \qquad \eta_i = y_i - p_i, \qquad \rho^2 = \sum_i \|\eta_i\|^2.$$

Write $\eta \in \mathbb{R}^{30}$ for the concatenation of the ten displacement vectors η_i. The quantity ρ is their total displacement. For alignment we choose the fixed candidate triple p_2, p_3, p_6 and the corresponding relabeled competitor triple. These need not be the anchor triple used during classification: (2) controls every pair, so it also controls these chosen triples. Aligning the anchor frames—the orthonormal bases obtained by Gram–Schmidt from these triples—controls their coordinates; the inner products with the anchor points then control the other points. The estimates in Appendix D.6 construct an alignment with total displacement at most 10000δ. A minimizing alignment is at least as good, so

$$\rho \le 10000\delta = 10^{-6}.$$

The global and local constants serve different steps. The windows have radius 10^{-10}, and the coordinate estimate loses a factor of 10^4. The resulting radius 10^{-6} is one thousand times smaller than the local energy estimate's permitted radius 10^{-3}.

Minimality of this orthogonal Procrustes alignment—least squares over orthogonal maps—also gives the skew conditions below. Here k, l index the three coordinates, and $\eta_{i,k}$ is coordinate k of the displacement η_i:

$$\sum_i (\eta_{i,k} p_{i,l} - \eta_{i,l} p_{i,k}) = 0 \quad (k, l \in \{1, 2, 3\}). \tag{3}$$

These equations make the displacement orthogonal to infinitesimal rotations. Each displacement also has the radial component $\langle p_i, \eta_i \rangle = -\|\eta_i\|^2/2$, by the exact unit-sphere identity. Thus the displacement itself need not lie in the tangent plane. Both properties enter the energy estimate below.

C.2 Stationarity and the full second variation

We establish stationarity and a positive quadratic bound for every perturbation after removing infinitesimal rotations. The candidate is stationary with respect to arbitrary tangent motions of all ten points. Write ∇_i for the gradient with respect to the three coordinates of point i. Since the energy is invariant under orthogonal maps and relabeling, its gradient transforms under the candidate's symmetries in the same way as the points do. A quarter-turn fixes each pole, so its gradient is vertical. The reflection in the vertical plane through p_2 fixes p_2, so its gradient lies in that plane. After removing the radial direction, the only possible gradient component there is along the motion of p_2 as the height changes.

All eight square points form one orbit under the fixed D_{4d} group, and changing height respects these same symmetries. Their contributions to the height derivative are therefore equal; the poles do not move. Thus $S(h_*)$ is eight times the contribution of p_2. Its vanishing makes that tangential component zero, and symmetry does the same at all eight square points. Every gradient is consequently radial.

More precisely, $\nabla_i \mathcal{E}(p) = \lambda_i p_i$, where $\lambda_i = -\frac{1}{2} \sum_{j \ne i} \|p_i - p_j\|^{-1}$. We call λ_i the radial coefficient; it is the spherical Lagrange multiplier. The gradient is the negative of the physical force. The corresponding formal lemmas are listed in Appendix D.7.

Let $H(h)$ be the 30×30 matrix given by the following blocks at $p(h)$, and abbreviate $H = H(h_*)$. At the stationary candidate, this is the constrained second variation. In the block formula at a general height, the radial coefficient means $\lambda_i(h) = -\frac{1}{2} \sum_{j \ne i} \|p_i(h) - p_j(h)\|^{-1}$; stationarity is asserted only at h_*. For a nonzero pair-difference vector d, the pair Hessian is

$$B(d) = \frac{3dd^{\mathsf{T}}}{\|d\|^5} - \frac{I_3}{\|d\|^3},$$

where the off-diagonal block of H for two different point indices is the negative of $B(p_i - p_j)$. The diagonal block at point i is the sum of $B(p_i - p_j)$ over the other nine points, minus the radial coefficient at that point times the three-by-three identity. These blocks specify the sphere-constrained second variation, including the radial correction. Similarly, let $R(h)$ have as its three columns the standard coordinate-axis infinitesimal rotations of $p(h)$, each with unit angular speed, and abbreviate $R = R(h_*)$.

Matrix bound. The exact matrix bound is

$$H + RR^{\mathsf{T}} - \frac{1}{20} I_{30} \succeq 0. \tag{4}$$

Rational interval arithmetic verifies the same inequality with $H(h)$ and $R(h)$ throughout $[a, b]$, hence at h_*. An invertible change of basis, applied by congruence to the matrix on the left of (4), transforms it into one with strictly positive diagonal entries, each greater than the sum of the absolute values of the other entries in its row. This strict diagonal dominance proves positive definiteness; the inverse change of basis gives (4). Appendix D.7 identifies the checked matrix calculation. For the concatenated displacement η, equation (3) gives $R^{\mathsf{T}}\eta = 0$, and (4) implies $\eta^{\mathsf{T}} H \eta \geq \rho^2/20$. The alignment allows all orthogonal maps; R removes their continuous rotation directions. The estimate covers every nearby point of $(S^2)^{10}$ after orthogonal alignment, including perturbations that break D_{4d} symmetry.

C.3 A finite-radius energy gap

To pass from the second variation to an energy inequality, we bound the higher-order error. For a nonzero pair-difference vector d and its perturbation w, put $\ell = \|d\| > 0$.

Pair remainder bound. For $\|w\| \leq \ell/200$, the algebraic pair estimate is

$$\frac{1}{\|d + w\|} \geq \frac{1}{\ell} - \frac{\langle d, w\rangle}{\ell^3} + \frac{3\langle d, w\rangle^2 - \ell^2\|w\|^2}{2\ell^5} - \frac{51}{50}\frac{\|w\|^3}{\ell^4}. \tag{5}$$

For this second-order Taylor bound with cubic error, put $z = (2\langle d, w\rangle + \|w\|^2)/\ell^2$. The proof bounds $(1 + z)^{-1/2}$ below by $1 - z/2 + 3z^2/8 - 5z^3/16$ on $-1 < z \leq 1$, then applies this bound to the normalized pair displacement. Every candidate pair distance is at least one. For $d = p_i - p_j$, $w = \eta_i - \eta_j$, the geometric bounds are

$$\|\eta_i - \eta_j\| \leq \tfrac{3}{2}\rho, \qquad \sum_{i<j} \|\eta_i - \eta_j\|^3 \leq 15\rho^3.$$

The first follows from $\|\eta_i - \eta_j\| \leq \sqrt{2}\rho \leq \frac{3}{2}\rho$. For the second, the identity for the sum of squared pair differences gives $\sum_{i<j} \|\eta_i - \eta_j\|^2 \leq 10\rho^2$. Multiplying this by the largest pair displacement gives the cubic bound. For $\rho \leq 1/1000$, these satisfy the hypothesis of (5). Summing the error over all 45 pairs bounds it by $(51/50)15\rho^3 = (153/10)\rho^3$.

The linear terms also have a quadratic expression on the sphere. By radial stationarity, their sum is the sum, over points, of the radial coefficient times $\langle p_i, \eta_i\rangle$. The exact sphere identity $\langle p_i, \eta_i\rangle = -\|\eta_i\|^2/2$ makes this minus one half of the sum of the radial coefficients times the squared displacements. Adding the quadratic terms from the pair Hessians therefore gives one half of the

constrained quadratic form $\eta^{\mathsf{T}} H \eta$: the radial correction is exactly the one subtracted in the diagonal blocks of H. Consequently, throughout $\rho \le 1/1000$, one obtains

$$\mathcal{E}(y) - \mathcal{E}(p) \;\ge\; \frac{1}{2}\eta^{\mathsf{T}} H \eta - \frac{153}{10}\rho^3 \;\ge\; \frac{\rho^2}{40} - \frac{153}{10}\rho^3 \;\ge\; \frac{\rho^2}{200}.$$

For every nonzero aligned displacement with $\rho \le 1/1000$, the final quantity is positive. Since $\mathcal{E}(y) = \mathcal{E}(x) \le \mathcal{E}(p)$, the inequality forces $\rho = 0$, hence $y = p$ exactly. Every competitor of energy at most $\mathcal{E}(p)$ is therefore congruent to p; all remaining competitors have strictly larger energy. Conversely, orthogonal transformations and permutations preserve energy. This proves both optimality and the equality case in Theorem 1. The candidate itself attains the minimum.

D Formal proof and certificates

This appendix identifies the formal statements, certificates and independent computations behind the preceding argument. The distribution contains a normal Lake project, rational certificates and a checker using Python's standard library. The Lean theorem begins with the definitions of unit sphere, injectivity, Coulomb energy and congruence; no numerical optimization output is treated as an axiom. The final theorem combines the root proof, pair separation, three-point bound, polynomial lower bound, the slack bound outside the seven windows, classification of the interval patterns and quantitative local energy inequality described above. The formal trust base is the Lean kernel and the three axioms printed below, together with the faithful match between its statement and Theorem 1. Paths beginning with `lean/` or `checks/` are relative to the bundle root. Other Lean source paths below are relative to `lean/ThomsonN10/`.

Checking artifacts. The project uses Lean 4.34.1. Its Mathlib revision is

`d13f23b723b8a846827a245b89c10fc7d3f11612`.

Both dependencies are pinned by the included toolchain and manifest. The build checks the theorem statements and the following `#print axioms` outputs. Any change to an axiom list makes the build fail:

```
'ThomsonN10.thomson_ten_complete' depends on axioms:
[propext, Classical.choice, Quot.sound]
'ThomsonN10.candidate_admissible' depends on axioms:
[propext, Classical.choice, Quot.sound]
'ThomsonN10.Cert.L1.Data.residual_agrees_JSON' depends on axioms:
[propext, Classical.choice, Quot.sound]
```

Neither `native_decide` nor `ofReduceBool` is used in the included Lean proof, including the decompressed certificate source. In particular, no native-computation axiom occurs in the printed transitive dependencies of the final theorem. These are the standard Lean axioms for propositional extensionality, classical choice and quotients.

The standard-library Python checker is an independent implementation that checks the same rational three-point certificate used by Lean. It expands the kernel and weighted-square polynomials from their definitions, checks the factor matrices and all 274 residual coefficients, bounds their total absolute error, and checks the candidate-energy upper bound over the entire height interval. The command `python3 -I checks/check.py` also runs eight corruption tests, all of which are rejected: it changes entries in each kind of factor matrix, a minorant coefficient, the candidate

upper bound, row exponents, row exponents together with the residual, a residual entry, and the residual norm. The Python implementation does not check the scalar minorant, the slack bound outside the windows, exhaustive passage from spherical configurations to interval patterns, window classification, full stationarity, orthogonal alignment or local Hessian argument; those obligations are proved by Lean. Its precise inputs and obligations are documented in `checks/README.md`. Independence here concerns the implementation of the shared certificate checks, not a second independent proof of every step of the global theorem.

D.1 The exact height interval

This subsection supplies the exact root interval and derivative formula. For the rational height interval used in Section 1, let A and D be the integers specified below, with D the denominator. Its endpoints a, b are defined by

$$a = A/D, \qquad b = (2A+1)/(2D),$$

$A = 513213568640798381215999065069135718731748351905698829816800875926548421101580455514066781,$
$D = 2^{283} \cdot 5^7.$

For the explicit stationarity equation, set

$$u_-(h) = 2 - \sqrt{2} + (2+\sqrt{2})h^2, \qquad u_+(h) = 2 + \sqrt{2} + (2-\sqrt{2})h^2.$$

The derivative of the family energy is

$$\begin{aligned} S(h) = \frac{(4\sqrt{2}+2)h}{(1-h^2)\sqrt{1-h^2}} - \frac{8(2+\sqrt{2})h}{u_-(h)\sqrt{u_-(h)}} - \frac{8(2-\sqrt{2})h}{u_+(h)\sqrt{u_+(h)}} \\ + \frac{8}{(2-2h)\sqrt{2-2h}} - \frac{8}{(2+2h)\sqrt{2+2h}} = F'(h). \end{aligned} \tag{6}$$

The root argument in Section 1 is formalized in `Param/Root.lean`. `Param/Final.lean` identifies those functions and endpoints with `stationarity`, `heightLower` and `heightUpper`. It proves `ParameterExistsUnique_holds`.

The type `CandidateParameter` consists exactly of reals in this interval satisfying $S(h) = 0$. Root uniqueness is asserted in this specified interval, not for all real heights.

D.2 Formal theorem statements

This subsection identifies the declarations that implement the theorem and their precise class of admissible competitors. The optimality and uniqueness assertions, together with parameter existence and uniqueness, form the declaration `ThomsonN10.thomson_ten_complete`. Its type is `ThomsonN10.Challenge.CompleteTarget`.

The other declaration `ThomsonN10.candidate_admissible` proves unit norm and injectivity for every `CandidateParameter`. Theorem 1 combines these two declarations. Each certificate assertion used in the proof is itself proved in Lean. In Lean, a configuration is a map from `Fin 10` to Euclidean three-space, and `IsSphereConfiguration` means exactly that each vector has norm one and the map is injective. The total Lean function uses the field convention $0^{-1} = 0$, so a collision is excluded by the injectivity hypothesis; it is not assigned infinite energy.

The following are the definitions in `Basic.lean`, with the mathematical glyphs typeset:

```
abbrev R3 := EuclideanSpace ℝ (Fin 3)
abbrev Configuration := Fin 10 → R3

def IsSphereConfiguration (x : Configuration) : Prop :=
  (∀ i, ‖x i‖ = 1) ∧ Function.Injective x

noncomputable def coulombEnergy (x : Configuration) : ℝ :=
  ∑ i : Fin 10, ∑ j ∈ Finset.Ioi i, (‖x i - x j‖)^-1

def Congruent (x y : Configuration) : Prop :=
  ∃ (g : R3 ≃ₗᵢ[ℝ] R3) (p : Equiv.Perm (Fin 10)), ∀ i, x i = g (y (p i))
```

Here the isometry-equivalence notation means a bijective real linear isometry, hence an element of $O(3)$. The challenge declarations in `lean/Challenge.lean` are:

```
def ParameterExistsUnique : Prop :=
  ∃! h : ℝ, (heightLower : ℝ) ≤ h ∧ h ≤ (heightUpper : ℝ) ∧ stationarity h = 0

def GlobalOptimality : Prop :=
  ∀ h : CandidateParameter, ∀ x : Configuration,
    IsSphereConfiguration x → coulombEnergy (candidate h) ≤ coulombEnergy x

def GlobalUniqueness : Prop :=
  ∀ h : CandidateParameter, ∀ x : Configuration,
    IsSphereConfiguration x →
    (coulombEnergy x = coulombEnergy (candidate h) ↔ Congruent x (candidate h))

def CompleteTarget : Prop := ParameterExistsUnique ∧ GlobalOptimality ∧ GlobalUniqueness
```

The first component specifies the height; the second is the lower bound; the third gives both directions of the equality case. Candidate validity is supplied by the separate theorem already identified above.

D.3 The separation polynomial

This records the quartic used to exclude too-close pairs before applying the three-point bound. Write P_k for the Legendre polynomial of degree k, normalized by $P_k(1) = 1$. The polynomial is

$$g(t) = \frac{17\,366\,680\,031}{21\,123\,877\,950} + \frac{833\,419\,493\,024}{1\,672\,307\,004\,375}P_1(t) + \frac{1\,760\,733\,078\,976}{7\,023\,689\,418\,375}P_2(t) + \frac{17\,766\,191\,104}{185\,811\,889\,375}P_3(t) + \frac{5\,504\,237\,568}{260\,136\,645\,125}P_4(t).$$

The nonnegative coefficients give the spherical-harmonic sum identity. The inequalities $g(t) \le f(t)$ on $[-1, 1)$ and $f(t) - g(t) > 51/1000$ on $[13/20, 1)$ follow from exact polynomial identities with nonnegative Bernstein coefficients. The identities and harmonic expansion are in `Cert/LP/Data.lean`; the scalar inequalities are in `Cert/LP/Scalar.lean`. The sum bound and separation conclusion are in `Cert/LP/Proof.lean`. The letter g here denotes the polynomial called `h` in those files.

D.4 The three-point kernel polynomials

This subsection specifies K and explains its nonnegative triple sum. Define polynomials $C_n(\alpha, \beta)$ by

$$C_0 = 1, \qquad C_1 = \alpha, \qquad C_{n+2} = 2\alpha C_{n+1} - \beta C_n, \qquad T_k(u, v, t) = C_k(t - uv, (1 - u^2)(1 - v^2)).$$

Here u and v are the inner products with the first, distinguished point, and t is the inner product of the other two points. Fixing that first point and projecting the others onto its perpendicular plane identifies the projected vectors with complex numbers. The recurrence expresses real parts of their powers. Therefore the quadratic sums of T_k, with weights given by products of real point weights, are squared absolute values of complex sums and are nonnegative. This is the spherical three-point positivity used in the proof.

For block k, let $J^{(k)}$ be its supplied factor matrix (the integer entries divided by 2^{200}, with unused entries set to zero in an eleven-by-eleven matrix). Its column polynomials and the polynomial with this distinguished first point are

$$\phi_{kr}(u) = \sum_{i=0}^{10} J^{(k)}_{ir} u^i, \qquad \mathcal{A}_k(u,v,t) = \sum_{r=0}^{10} \phi_{kr}(u)\phi_{kr}(v)T_k(u,v,t).$$

Let $\overline{\mathcal{A}}_k$ be the average of $\mathcal{A}_k$ over all six permutations of its three arguments. The repeated-index correction and full kernel are

$$K_k(u,v,t) = 8\overline{\mathcal{A}}_k(u,v,t) + \overline{\mathcal{A}}_k(u,u,1) + \overline{\mathcal{A}}_k(v,v,1) + \overline{\mathcal{A}}_k(t,t,1) + \tfrac{1}{9}\overline{\mathcal{A}}_k(1,1,1),$$
$$K(u,v,t) = \sum_{k=0}^{9} K_k(u,v,t).$$

The coefficients account for repeated indices in the full triple sum. There are $10 \cdot 9 \cdot 8 = 720$ ordered triples with distinct indices, $10 \cdot 9 = 90$ for each of the three ways exactly two indices coincide, and ten with all indices equal. Divide the full sum by 90: the distinct triple average has coefficient $720/90 = 8$, each repeated-pair average has coefficient one, and the constant has coefficient $10/90 = 1/9$. Averaging K_k over distinct triples gives exactly this normalized full sum. Its nonnegativity therefore gives the required distinct-triple positivity of K. The recurrence and projection argument are in `Cert/SDP/CircleKernel.lean` and `Cert/SDP/ProjectionComplex.lean`; the symmetrization and correction are in `Cert/SDP/FactoredKernels.lean`.

D.5 Certificate data and finite classification

This subsection specifies the rational data and weights used in Lemma 1, then describes the finite classification. The coefficients c_j are `hCoeff` in `lean/ThomsonN10/Data.lean`; `certU` there is the exact rational

$$U = \frac{52\,574\,110\,779\,609\,793\,015\,977\,559\,478\,279\,245\,057\,879\,406\,410\,836\,764\,596\,571\,436}{2^{200}}.$$

These are fixed data, not optimization variables in the theorem. The candidate upper bound is verified by `Cert/CandidateUpper.lean`.

Write $\Delta = 1+2uvt-u^2-v^2-t^2$ for the Gram determinant. Before averaging over permutations of u, v, t, the six weights multiplying polynomial squares are 1, $1+u$, $13/20-u$, Δ, $(1+u)(13/20-u)$ and $\Delta(1+u)(13/20-u)$. Each is used for two blocks, one symmetric and one antisymmetric under interchange of v, t; after squaring and permutation averaging all twelve blocks are nonnegative on the capped Gram domain. These weights are `domainWeight` in `Cert/SDP/CertificateBound.lean`.

Averaging the residual identity over the $10 \cdot 9 \cdot 8$ ordered triples of distinct indices turns its pair term into $\sum_{i<j} m(\langle x_i, x_j\rangle)/45$. This is because each of the 45 unordered pairs occurs equally often in the three positions. The polynomial shift 2^{-130} is included in m, so the reference constant is $(U - 45 \cdot 2^{-130})/45$. Bounding the averaged residual by $\|\mathcal{R}\|_1$ and multiplying by 45 gives

$U - L$. In particular, the shift is included once in the defined loss L. The displayed 10^{-30} bound is conservative: most of L comes from the downward minorant shift, while the residual norm is about 10^{-56}. No exact-sharpness assertion is inferred from these small numbers.

The certificate stores 20 matrices of factors: eight nonzero matrices for the three-point kernels (called moment blocks in the data), and twelve for the weighted squares (called sum-of-squares blocks). It has 42,950 integer entries and 274 residual coefficients; two zero moment blocks are omitted. Its compressed JSON file is 1,246,399 bytes and expands to 2,670,249 bytes. Expanding these factors and bounding the coefficient sum are exact rational calculations. The identity is `Cert.L1.l1_hidentity`; its summation is `Cert.SDP.factored_certificate_pair_bound`.

In the data, a *norm test* is the four-by-four Gram-determinant interval test for one proposed node; an *extension record* stores one retained node. A *pair-color test* checks the cross-Gram determinant for two nodes and a proposed pair color. A *seven-clique witness* is the color-preserving candidate labeling for seven compatible nodes and the three anchor points.

The finite certificate treats 84 sorted anchor-color triples, of which 43 are eligible. It contains 14,749 norm tests, 79 extension records and 1,960 pair-color tests. Its nine seven-clique witnesses each give a bijective relabeling to the same reference interval pattern. The count is of witnesses in the enumeration, not of incongruent ten-point configurations.

The interval-pattern conclusion is the theorem `Windows.WindowClassification_holds`.

Scalar checks and data formats. The minorant and the slack bound outside the windows are certified by subdividing rational intervals and checking nonnegative Bernstein coefficients of integer polynomials. The substitution $t = 1 - 2y^2$, with $y = \sqrt{(1-t)/2} > 0$, turns the reciprocal-square-root inequalities into polynomial inequalities after multiplication by positive denominators. The soundness proof is in `Scalar/Checker.lean`; `Scalar/Bridge.lean` carries out the substitution and `Scalar/Exterior.lean` proves the slack bound outside the windows.

The compressed JSON file is input to the independent Python checker. Lean uses the corresponding coefficient tables as literal Lean data; it does not parse that JSON file during proof checking. The theorem `residual_agrees_JSON` equates the residual computed from those factors with the literal polynomial `JSONResidual` exported from the supplied table. The Python checker separately verifies the external JSON residual and its coefficient norm. The compressed file `AmbientKernelData.lean.gz` contains Lean source for the Hessian certificate, not JSON. The build expands that source before importing it; all its assertions are then kernel-checked. No proof generator is run as a trusted build step.

D.6 Conditioning of the anchor alignment

A quantitative Gram-stability argument turns (2) into an orthogonal alignment. It uses the candidate anchor points p_2, p_3, p_6: the absolute values of the inner products of pairs $(2,3)$, $(2,6)$ and $(3,6)$ are at most $19/100$, $41/100$ and $41/100$, respectively. The corresponding pairs of relabeled competitor anchor points consequently satisfy the absolute inner-product bounds $1/5, 5/12, 5/12$. In Gram–Schmidt, each residual is the component of an anchor point orthogonal to the span of the preceding anchor points. The lengths of the two residuals are at least $2/3$ and $1/2$. Thus the divisions used to recover coordinates from inner products have denominators bounded away from zero.

In the resulting orthonormal frames, the coordinate differences for each point are at most δ, 22δ and 646δ, respectively. These scalar estimates follow by applying the product, square-root and quotient difference bounds to the Gram–Schmidt formulas; they are proved in `Local/GramStability.lean`. Relaxing each to 650δ and summing the 30 coordinate squares gives $\rho^2 \leq$

$30(650\delta)^2 \le (10000\delta)^2$ for the constructed alignment. `Local/GramAlignment.lean` transfers this bound to the minimizing alignment.

D.7 Local inequalities in Lean

This subsection identifies the formal inequalities used for full stationarity and the finite-radius energy gap. The interval Hessian calculation encloses the arithmetic and square-root operations over the full height interval. Its change-of-basis matrix is upper triangular with all diagonal entries equal to one, hence invertible. Interval bounds on the transformed matrix certify strict diagonal dominance. In `Local/Certificates/AmbientKernelBounds.lean`, the theorem `flattened_posDef` proves positive definiteness.

The theorem that proves (4) for the explicitly defined Hessian is listed below its source file. The last line is the Lean namespace prefix for both this theorem and `flattened_posDef`:

```
Local/Certificates/AmbientHessian.lean
candidate_augmented_hessian_posSemidef
ThomsonN10.Local.Certificates.AmbientKernel
```

The formal coordinate calculation rewrites the height equation as a relation among the seven inverse-cube distance weights, one for each candidate inner-product class, and checks the gradient at every point. The stationarity calculation in Section C has the following formal correspondence. In `Local/LocalStationarity.lean`, `candidate_force_relation` converts that derivative equation to a linear relation among the seven inverse-cube distance weights. The identity `orbit_force_balance` checks all force components and uses this relation to make each point's force radial; `candidate_force_eq_radialCoefficient` identifies the multiplier with the exact spherical radial coefficient. This proves stationarity for arbitrary tangent perturbations at all ten points.

The pair remainder inequality (5) is `pair_potential_algebraic_lower`. The local estimate uses the exact unit-sphere identity $\langle p_i, y_i - p_i\rangle = -\|y_i - p_i\|^2/2$. Its final conclusion is $\mathcal{E}(y) - \mathcal{E}(p) \ge \rho^2/200$ for aligned displacement norm $\rho \le 1/1000$.

D.8 The separate energy calculation

This subsection explains how to check the separate energy enclosure in Section A. This calculation is checked only by Python. It is computed by rational interval evaluation of the candidate's 45 pair energies over $[a, b]$. Square roots are enclosed using integer square roots on a dyadic grid of spacing 2^{-420}, rounding outward. Reciprocal and summation operations preserve the rational bounds. The script `checks/energy_enclosure.py` performs this computation and prints the two endpoints in the energy remark in Section A. Lean proves that the height interval $[a, b]$ contains its unique stationary root; Python uses the interval as input and does not prove root existence or uniqueness.

D.9 Code and cutoff provenance

The original code attribution, historical comparisons and cutoff account are retained in `checks/context.md`; the file-level credits and full MIT notice remain in `THIRD_PARTY.md`.

References

[1] J. J. Thomson, On the structure of the atom: an investigation of the stability and periods of oscillation of a number of corpuscles arranged at equal intervals around the circumference of a circle; with application of the results to the theory of atomic structure, *Philosophical Magazine* 7 (1904), 237–265. doi:10.1080/14786440409463107.

[2] A. Karthigeyan (director and publisher), *The Nine-Point Coulomb Thomson Problem*, MIT-licensed Lean project, revision `168ff0c`, October 6, 2026. https://github.com/aravindkarthigeyan/thomson-n9-lean/tree/168ff0c98cab4dfda0460ba55f8d169070ac385e.

[3] C. Bachoc and F. Vallentin, New upper bounds for kissing numbers from semidefinite programming, *Journal of the American Mathematical Society* 21 (2008), 909–924. doi:10.1090/S0894-0347-07-00589-9.

arXiv version 4, October 3, 2007. https://arxiv.org/abs/math/0608426v4.

[4] H. Cohn and J. Woo, Three-point bounds for energy minimization, *Journal of the American Mathematical Society* 25 (2012), 929–958. arXiv:1103.0485v2. Version 2, June 24, 2013.